

Tarea 18

Masa molar

1) Datos

$$^{35}_{17}\text{Cl} = 75.53\%, 34.96 \text{uma} \quad | \quad ^{37}_{17}\text{Cl} = 24.47\% = 36.95 \text{uma}$$

$$\frac{75.53}{100} = 0.7553$$

$$\frac{24.47}{100} = 0.2447$$

$$34.96(0.7553) + 36.95(0.2447) = 35.44 \rightarrow \text{masa atómica promedio}$$

2) Datos

$$^{63}_{29}\text{Cu} = 69.17\%, 62.92 \text{uma}$$

$$^{65}_{29}\text{Cu} = 30.83\%, 64.9278$$

$$0.6917(62.92) + 0.3083(64.9278) = 63.5456$$

MAP masa atómica

$$\text{Número de avogadro} = 6.022 \times 10^{23}$$

$$1) \frac{6.46 \text{ g He}}{4.003 \text{ g He}} \left| \frac{1 \text{ mol He}}{4.003 \text{ g He}} \right| = 1.61 \rightarrow \text{moles}$$

$$2) \text{CO} = 6 \times 10^9 \times \frac{1 \text{ mol}}{6.022 \times 10^{23} \text{ CO}} = 9.96 \times 10^{-15}$$

$$3) 15.3 \text{ mol Au} \times \frac{196.96 \text{ g}}{1 \text{ mol Au}} = 3,013.49 \text{ g}$$

Massa molecular

H_2O , massa molecular

$$\text{H} = 1.00794 \times 2 = 2.01588$$

$$\text{O} = 15.9994$$

$$18.01528 \rightarrow \text{massa molecular}$$

CH_3OH

$$\text{C} = 12.011$$

$$\text{H} = 4.03176$$

$$\text{O} = 15.9994$$

$$32.04 \rightarrow \text{massa molecular}$$

1) Dato: ¿cuántas moles hay en 6.67g de CH_4 ?

$$6.67 \text{ g}$$

CH_4

$$\text{C} = 12.011$$

$$\text{H} = 4.03176$$

$$16.04276 \text{ g}$$

$$\frac{6.67 \text{ g}}{16.04 \text{ g}} \times \frac{1 \text{ mol}}{1} = 0.42$$

2) ¿Cuántos moles de átomos de calcio (Ca) hay en 77.4g de Ca?

$$\frac{77.4g \text{ Ca}}{40.08g} \times \frac{1 \text{ mol}}{1} = 1.93 \text{ mol}$$

3) $[(\text{CH}_3)_2\text{SO}] = 78.13$

$$\text{C}_2 = 24.022$$

$$\text{H}_6 = 6.05$$

$$\text{S} = 32.07$$

$$\text{O} = 15.99$$

$$\frac{7.74 \times 10^3 \text{ g}}{78.13 \text{ g}} \times \frac{1 \text{ mol}}{1} \times \frac{2 \text{ mol C}}{1 \text{ mol } [(\text{CH}_3)_2\text{SO}]} \times \frac{6.022 \times 10^{23} \text{ atom C}}{1 \text{ mol C}} = 1.10 \times 10^{26} \text{ atom C}$$

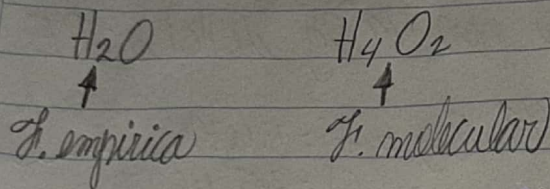
$$\frac{7.74 \times 10^3 \text{ g}}{78.13 \text{ g}} \times \frac{1 \text{ mol}}{1} \times \frac{6 \text{ mol H}}{1 \text{ mol } [(\text{CH}_3)_2\text{SO}]} \times \frac{6.022 \times 10^{23} \text{ atom H}}{1 \text{ mol H}} = 3.30 \times 10^{26} \text{ atom H}$$

$$\frac{7.74 \times 10^3 \text{ g}}{78.13 \text{ g}} \times \frac{1 \text{ mol}}{1} \times \frac{1 \text{ mol S}}{1 \text{ mol } [(\text{CH}_3)_2\text{SO}]} \times \frac{6.022 \times 10^{23} \text{ atom S}}{1 \text{ mol S}} = 5.50 \times 10^{25} \text{ atom S}$$

Composición porcentual de los compuestos

$$\begin{aligned} \text{CHCl}_3 &= 119.38 \\ \text{C} &= 12.011 \div 119.38 = 0.10 \times 100 = 10\% \\ \text{H} &= 1.00794 \div 119.38 = 0.84\% \\ \text{Cl}_3 &= 70.36 \div 119.38 = 0.59 \times 100 = 59\% \end{aligned}$$

Determinación experimental de fórmulas empíricas



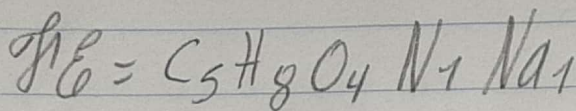
$$1) \text{C} = 35.51\% - 35.51\text{g} \times \frac{1\text{mol}}{12.011\text{g}} = 3\text{mol} \quad 5$$

$$\text{H} = 4.77\% - 4.77\text{g} \times \frac{1\text{mol}}{1.00794\text{g}} = 4.73\text{mol} \quad 8$$

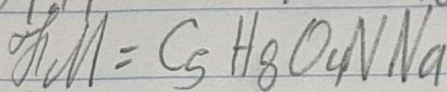
$$\text{O} = 37.85\% - 37.85\text{g} \times \frac{1\text{mol}}{15.9994\text{g}} = 2.37\text{mol} \quad 4$$

$$\text{N} = 8.29\% - 8.29\text{g} \times \frac{1\text{mol}}{14.0064\text{g}} = 0.59 \quad 1$$

$$\text{Na} = 13.60\% - 13.60\text{g} \times \frac{1\text{mol}}{22.98977\text{g}} = 0.59 \quad 1$$



$$m = 169.11 \quad 169 \div 169.11 = 1$$



2) gases

$$1) \text{Vol}_1 = 3.20 \text{ L} \\ \text{Temp}_1 = 125^\circ\text{C} = 398 \text{ K}$$

$$\text{Vol}_2 = 7.54 \text{ L} \\ \text{Temp}_2 = ?$$

$$T_2 = \frac{V_1 T_1}{V_2} \quad T_2 = \frac{3.20 \text{ L} \cdot 398 \text{ K}}{7.54 \text{ L}} = 827^\circ\text{K}$$

2) Volumen en litros ocupado por 49.8g de HCl a TPE

$$V = nRT/P$$

$$\text{HCl} = 49.8 \text{ g} = \frac{49.8}{7.00794} = 7.106 \text{ mol}$$

$$\text{Cl} = \frac{35.4527}{36.46}$$

$$V = \frac{7.106 \text{ mol} \cdot (0.0821 \text{ L} \cdot \text{atm} / \text{mol} \cdot \text{K}) \cdot 273.15 \text{ K}}{1 \text{ atm}}$$

$$V = 30.69 \text{ L}$$

$$49.8 \text{ g} \left| \frac{1 \text{ mol}}{36.46 \text{ g}} \right. = 1.36 \text{ mol}$$